

Homework 5 File Submission

A. Convert the decimal expansion of 321 to a binary expansion. Show all work.

$$321 \div 2 = 160 \text{ r } 1$$

$$160 \div 2 = 80 \text{ r } 0$$

$$80 \div 2 = 40 \text{ r } 0$$

$$40 \div 2 = 20 \text{ r } 0$$

$$20 \div 2 = 10 \text{ r } 0$$

$$10 \div 2 = 5 \text{ r } 0$$

$$5 \div 2 = 2 \text{ r } 1$$

$$2 \div 2 = 1 \text{ r } 0$$

$$1 \div 2 = 0 \text{ r } 1$$

$$321 = (1\ 0100\ 0001)_2$$

C. Convert the binary expansion of $(1010\ 1010\ 1010)_2$ to an octal expansion. Show all work.

1. Convert to decimal expansion:

$$\begin{aligned} (1010\ 1010\ 1010)_2 &= 2^{11} + 2^9 + 2^7 + 2^5 + 2^3 + 2^1 \\ &= 2048 + 512 + 128 + 32 + 8 + 2 \\ &= 2730 \end{aligned}$$

2. Now convert decimal to octal expansion:

$$2730 \div 8 = 341 \text{ r } 2$$

$$341 \div 8 = 42 \text{ r } 5$$

$$42 \div 8 = 5 \text{ r } 2$$

$$5 \div 8 = 0 \text{ r } 5$$

$$(1010\ 1010\ 1010)_2 = (5252)_8$$

B. Convert the octal expansion of $(1604)_8$ to a binary expansion. Show all work.

1. Convert to decimal expansion:

$$(1604)_8 = 1 \cdot 8^3 + 6 \cdot 8^2 + 0 \cdot 8^1 + 4 \cdot 8^0 = 512 + 384 + 4 = 900$$

2. Convert to binary expansion:

$$900 \div 2 = 450 \text{ r } 0$$

$$450 \div 2 = 225 \text{ r } 0$$

$$225 \div 2 = 112 \text{ r } 1$$

$$112 \div 2 = 56 \text{ r } 0$$

$$56 \div 2 = 28 \text{ r } 0$$

$$28 \div 2 = 14 \text{ r } 0$$

$$14 \div 2 = 7 \text{ r } 0$$

$$7 \div 2 = 3 \text{ r } 1$$

$$3 \div 2 = 1 \text{ r } 1$$

$$1 \div 2 = 0 \text{ r } 1$$

$$(1604)_8 = (11\ 1000\ 0100)_2$$

D. Find the sum and product of $(2112)_3$, $(12021)_3$. Express your answer as a base 3 expansion. Show all work.

$$\text{Let } A = (2112)_3 \text{ and } B = (12021)_3$$

Convert to decimal:

$$A = 2 \cdot 3^3 + 1 \cdot 3^2 + 1 \cdot 3 + 2 = 54 + 9 + 3 + 2 = 68$$

$$B = 1 \cdot 3^4 + 2 \cdot 3^3 + 0 \cdot 3^2 + 2 \cdot 3 + 1 = 81 + 54 + 0 + 6 + 1 = 142$$

$$\text{Sum: } A + B = 68 + 142 = 210$$

$$210 \div 3 = 70 \text{ r } 0$$

$$70 \div 3 = 23 \text{ r } 1$$

$$23 \div 3 = 7 \text{ r } 2$$

$$7 \div 3 = 2 \text{ r } 1$$

$$2 \div 3 = 0 \text{ r } 2$$

$$A + B = (21210)_3$$

$$\text{Product: } A \cdot B = 68 \cdot 142 = 9656$$

$$9656 \div 3 = 3218 \text{ r } 2$$

$$3218 \div 3 = 1072 \text{ r } 2$$

$$1072 \div 3 = 357 \text{ r } 1$$

$$357 \div 3 = 119 \text{ r } 0$$

$$119 \div 3 = 39 \text{ r } 2$$

$$39 \div 3 = 13 \text{ r } 0$$

$$13 \div 3 = 4 \text{ r } 1$$

$$4 \div 3 = 1 \text{ r } 1$$

$$1 \div 3 = 0 \text{ r } 1$$

$$A \cdot B = (111020122)_3$$